

3590 883

COPYRIGHT RESERVED

BCA (Sem-V) – Theory

of Computation (502) Core-XII

2023

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

Compulsory

1. Choose the correct alternative for each of the
following questions : 2×10=20

(a) The class of CFG is not closed under :

- i. Concatenation
- ~~ii. Intersection~~
- iii. Union
- iv. Repeated concatenation

(b) Minimum number of states require to accept string ends with 101.

- ~~i. 3~~
- ii. 2
- iii. 1
- ~~iv. Can't be represented~~

(c) Automata accepting the regular expression of any number of a's is :

- ~~i. a^*~~
- ii. ab^*
- iii. $(a/b)^*$
- iv. $a^*b^*c^*$

(d) The regular sets are closed under :

- i. Union
- ii. Concatenation
- iii. Kleen Closure
- ~~iv. All of the above~~

(e) A language is regular if and only if :

- ~~i. Accepted by DFA~~
- ii. Accepted by PDA
- iii. Accepted by LBA
- iv. Accepted by Turing machine

(f) Grammar that produce more than one parse tree for same sentence is :

- ~~i. Ambiguous~~
- ii. Unambiguous
- iii. Complement
- iv. Concatenation

(g) The logic of pumping lemma is a good example of :

- i. Pigeon hole principle
- ii. Divide and conquer technique
- iii. Recursion
- iv. Iteration

(h) Pumping lemma is generally used for proving that :

- i. Given grammar is regular
- ☒ ii. Given grammar is not regular
- iii. Expression are equivalent or not
- iv. None of these

(i) What is the minimum number of states required by a Turing machine to accept strings which consist of even number of 1's ?

i. 5

ii. 1

☒ iii. 2

iv. 4

(j) Which of the following machine is specific for context free grammar ?

i. Finite state automata

☒ ii. Push down automata

iii. Linear bounded automata

iv. Turing machine

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

2. What is language ? Explain with example.

3. What is finite automata ? Explain with example.
4. What is difference between DFA and NFA ?
5. State and prove the pumping lemma for regular language $L = \{0^n | 0^n | n \geq 1\}$ is not a regular language.
6. Construct context free grammar to language having any no 7 a's over the set $\Sigma = \{a\}$.
7. What is deterministic finite automata ? Explain with example.
8. What is ambiguity grammar ? Explain.

Section-C

Long Answer type Questions

Answer any two questions of the following :

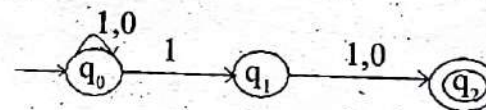
15×2=30

S219/4/2

(6)

Contd.

9. Define push down automata. Construct push down automata for the language $L = \{WW^R | W \in (a+b)^*\}$.
10. Prove that $L = \{a^i b^i c^i | i \geq 1\}$ is not context free language.
11. Define Turing machine. Design a turing machine of the language $L = \{a^n b^n | n > 0\}$.
12. Convert the following NFA to equivalent DFA.



_____ x _____

S219/4/2

(7)

(P-600)